

CHIA MIN YAN



Phone: (60)
192508182

Email: chiamy94@gmail.com

Address: 51A, Jalan Serunai 4, Taman Desa
Utama, 41200 Klang, Selangor, Malaysia.

EDUCATION

Doctor of Philosophy (Engineering)

June 2019 – June 2022

Thesis: Robust Data Fusion Techniques Integrated Machine Learning Models for Estimating Reference Evapotranspiration

Master of Engineering Science

Jan 2017 – May 2019

Dissertation: Development of High Performance Sulfonated Poly (Ether Ether Ketone) Based Composite Proton Exchange Membranes

Bachelor of Engineering (Honours) Chemical Engineering

Jan 2013 – Oct 2016

CGPA: 3.8758 (Distinction)

WORKING EXPERIENCE

Data Scientist | PAIDChain Sdn. Bhd.

July 2022 – present

Data Scientist | JuiceUp EV Grid Sdn. Bhd. (100 % subsidiary of PAIDChain Sdn. Bhd.)

Jan 2024 – present

- Research on and design statistical as well as machine learning models to meet the business requirements in the fintech industry (fraud detection, merchant monitoring, website monitoring, credit scoring)
- Manage several microservices on the backend (databases and APIs creation) and lead frontend developments (web portals and dashboards)
- Initiate the setup of artificial intelligence for business use case, including the implementation of LLMs and agentic frameworks
- Technology stack: Node.js, Typescript, PostgreSQL, Angular, Git, Docker, Python

Production Engineer | Maxter Glove Manufacturing Sdn. Bhd.

Jan 2019 – Apr 2019

- Monitor the quality of the produced gloves to meet the customers' requirements
- Documentation of regular maintenance for the production lines

Production Engineer (Intern) | Maxter Glove Manufacturing Sdn. Bhd.

Oct 2015 – Dec 2015

- Ensure the compliance of wastewater quality to the regulated standards
- Optimise the process parameters of wastewater treatment plant for cost saving purpose

RESEARCH EXPERIENCE

Research Assistant | Universiti Tunku Abdul Rahman

June 2019 – May 2022

- Assist in drafting research proposals and projects
- Manuscript writing for international journal publications
- Assist in guiding final year project students

Research Assistant | Universiti Tunku Abdul Rahman

Nov 2016 – Nov 2018

- Perform laboratory demonstrations
- Manuscript writing for international journal publications
- Assist in guiding final year project students

PROFESSIONAL AFFILIATIONS

Graduate Member of Board of Engineers Malaysia (BEM)

2018–present

Graduate Member of The Institution of Engineers, Malaysia (IEM)

2017–present

Associate Member of Institution of Chemical Engineers (ICChemE)

2017–present

Graduate Technologist of Malaysia Board of Technologists (MBOT)

2022–present

PUBLICATIONS

JOURNAL ARTICLES

Wong, Y.J.*, Yeganeh, A., **Chia, M.Y.***, Shiu, H.Y., Ooi, M.C.G., Chang, J.H.H., Shimizu, Y., Ryosuke, H., Try, S. and Elbeltagi, A., 2023. Quantification of COVID-19 impacts on NO₂ and O₃: Systematic model selection and hyperparameter optimization on AI-based meteorological-normalization methods. *Atmospheric Environment*, p.119677 [Web of Science: Q1].

Wai, K.P., **Chia, M.Y.**, Koo, C.H.*, Huang, Y.F. and Chong, W.C., 2022. Applications of deep learning in water quality management: A state-of-the-art review. *Journal of Hydrology*, 613, p.128332 [Web of Science: Q1].

Chia, M. Y., Huang, Y. F.*, Koo, C. H., Ng, J. L., Ahmed, A. N. and El-Shafie, A., 2022. Long-term forecasting of monthly mean reference evapotranspiration using deep neural network: A comparison of training strategies and approaches. *Applied Soft Computing*, pp. 109221 [Web of Science: Q1].

Chia, M.Y., Huang, Y.F.* and Koo, C.H., 2022. Resolving data-hungry nature of machine learning reference evapotranspiration estimating models using inter-model ensembles with various data management schemes. *Agricultural Water Management*, 261, p.107343 [Web of Science: Q1].

Chia, S.L., **Chia, M.Y.**, Koo, C.H.* and Huang, Y.F., 2021. Integration of advanced optimization algorithms into least-square support vector machine (LSSVM) for water quality index prediction. *Water Supply* [Web of Science: Q4].

Chia, M.Y., Huang, Y.F.* and Koo, C.H., 2021. Improving reference evapotranspiration estimation using novel inter-model ensemble approaches. *Computers and Electronics in Agriculture*, 187, p.106227 [Web of Science: Q1].

Chia, M.Y., Huang, Y.F.* and Koo, C.H., 2021. Swarm-based optimization as stochastic training strategy for estimation of reference evapotranspiration using extreme learning machine. *Agricultural Water Management*, 243, p.106447 [Web of Science: Q1].

Chia, M.Y., Thiam, H.S.*, Leong, L.K., Koo, C.H. and Saw, L.H., 2020. Study on improvement of the selectivity of proton exchange membrane via incorporation of silicotungstic acid-doped silica into SPEEK. *International Journal of Hydrogen Energy*, 45(42), pp.22315-22323 [Web of Science: Q2].

Chia, M.Y., Huang, Y.F.* and Koo, C.H., 2020. Support vector machine enhanced empirical reference evapotranspiration estimation with limited meteorological parameters. *Computers and Electronics in Agriculture*, 175, p.105577 [Web of Science: Q1].

Chia, M.Y., Huang, Y.F.*, Koo, C.H. and Fung, K.F., 2020. Recent advances in evapotranspiration estimation using artificial intelligence approaches with a focus on hybridization techniques—a review. *Agronomy*, 10(1), p.101 [Web of Science: Q1].

CONFERENCE ARTICLES

Chia, M.Y., Huang, Y.F.*, Koo, C.H. and Chong, Z.R., 2022. Long-term forecasting of climatic parameters using parametric and non-parametric stochastic modelling. In *E3S Web of Conferences* (Vol. 347, p. 05013). EDP Sciences [Scopus].

Chia, M.Y., Huang, Y.F.*, Koo, C.H. and Lai, L.S., 2022. Feasibility of bootstrap aggregating to enhance extreme learning machine for reference evapotranspiration estimation. In *E3S Web of Conferences* (Vol. 347, p. 04003). EDP Sciences [Scopus].

Chia, M.Y., Huang, Y.F.* and Koo, C.H., 2021, April. ANN-Based Reference Evapotranspiration Estimation: Effects of Data Normalization and Parameters Selection. In *International Conference on Emerging Technologies and Intelligent Systems* (pp. 3-12). Springer, Cham [Scopus].

Chia, M.Y., Huang, Y.F.*, Koo, C.H. and Tan, Y.Z., 2021, April. Trend Analysis of the Effects of Climate Change on the Pan Evaporation Rate in Sabah, Malaysia. In *International Conference on Emerging Technologies and Intelligent Systems* (pp. 13-22). Springer, Cham [Scopus].

Chia, M.Y., Huang, Y.F.* and Koo, C.H., 2020, December. Reference evapotranspiration estimation using adaptive neuro-fuzzy inference system with limited meteorological data. In *IOP Conference Series: Earth and Environmental Science* (Vol. 612, No. 1, p. 012017). IOP Publishing [Scopus].

Chia, M.Y., Thiam, H.S.*, Leong, L.K., Koo, C.H. and Saw, L.H., 2019, June. Effect of coupling agents on the transport properties of sulfonated poly (ether ether ketone) based composite membrane. In *IOP Conference Series: Earth and Environmental Science* (Vol. 268, No. 1, p. 012131). IOP Publishing [Scopus].

Chia, M.Y., Thiam, H.S.*, Leong, L.K. and Koo, C.H., 2018, August. Effect of Filler Content on Transport Properties of Sulfonated Polyether Ether Ketone (SPEEK) Composite Proton Exchange Membranes. In *IOP Conference Series: Materials Science and Engineering* (Vol. 409, No. 1, p. 012003). IOP Publishing [Scopus].

Thiam, H.S.*, **Chia, M.Y.**, Cheah, Q.R., Koo, C.C.H., Lai, S.O. and Chong, K.C., 2017, April. Proton conductivity and methanol permeability of Nafion-SiO₂/SiWA composite membranes. In *AIP Conference Proceedings* (Vol. 1828, No. 1, p. 020007). AIP Publishing LLC [Scopus].

CONFERENCES ATTENDED

2nd International Conference on Civil and Environmental Engineering
6–7 January 2022, Online Conference

International Conference on Emerging Technologies and Intelligent Systems 2021
25–26 June 2021, Online Conference

International Conference on Sustainable Energy and Green Technology 2018
11–14 December 2018, Kuala Lumpur, Malaysia

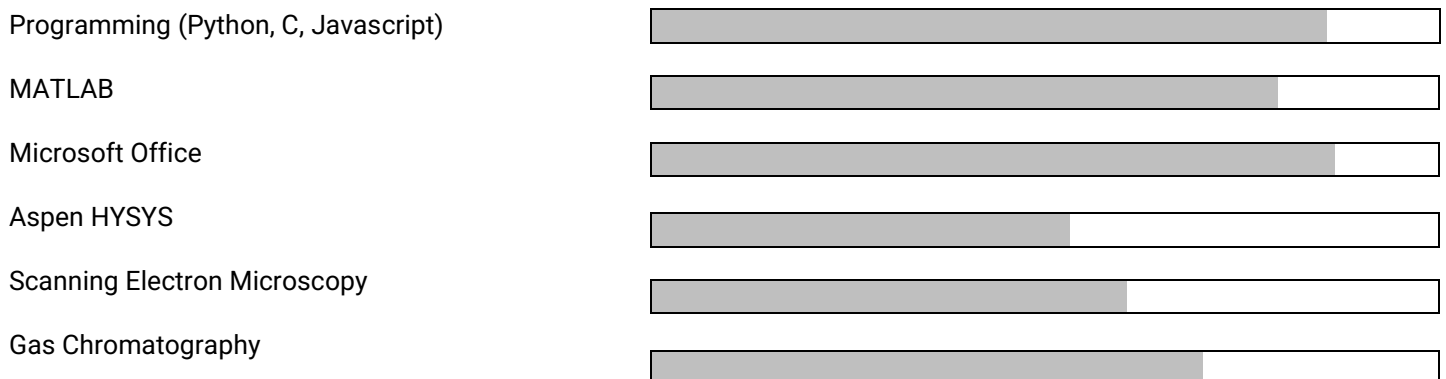
RESEARCH INTERESTS

- Environmental Modelling
- Agentic AI for Decision Process
- Artificial Intelligence for Engineering Applications
- Large Language Model for Engineering Applications
- Fuel Cell Membrane
- Fuel Cell Engineering

LANGUAGE PROFICIENCY



SKILLS



EXTRACURRICULAR ACTIVITIES

Member of Universiti Tunku Abdul Rahman (Sungai Long Campus) Chinese Debate Team	2018–2022
Participation in the 17 th National Varsity Chinese Debate Competition	2020
Participation in the 2018 Nan Ying Cup International Intervarsity Chinese Debate Competition	2018
Participation in the UCSI University Chinese Debate Tournament	2018
Participation in the 16 th National Varsity Chinese Debate Competition	2018

REFERENCES

Lee Yuen Fai

Senior Software Engineer
PAIDChain Sdn. Bhd. | Juice Up EV Grid Sdn. Bhd.
(60) 122386025
bryan.lee@paidchain.my

Professor Ir. Dr. Huang Yuk Feng

Professor
Universiti Tunku Abdul Rahman, Malaysia
(60) 129272332
huangyf@utar.edu.my